(U) Accreditation Report

(U) T-150/TRV-150 Heavy-Lift UAS
AFSIM Simulation Model v2.0
(U) AFSIM Project 005



Version: 1.0

Date: March 5, 2026
Classification: UNCLASSIFIED

Table 1: (U) Document Information

UNCLASSIFIED	
Field	Value
Document Version	1.0
Date	March 5, 2026
Classification	UNCLASSIFIED
	UNCLASSIFIED

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Table 1: (U) Document Information (Continued)

UNCLASSIFIED	
Field	Value
Sponsoring Organization	[PLACEHOLDER - Organization Name]
Accreditation Authority	[PLACEHOLDER - Name/Title]
Document Preparer	[PLACEHOLDER - Organization/Contract]
Distribution Statement	[PLACEHOLDER - Distribution Statement]
M&S Name and Version	T-150 UAS AFSIM Plugin Suite v2.0
Accreditation Decision	CONDITIONAL ACCREDITATION
	UNCLASSIFIED

(U) This document is prepared in accordance with:

- (U) **DoDI 5000.61**, "DoD Modeling and Simulation Verification, Validation, and Accreditation," September 17, 2024 [1]
- (U) **MIL-STD-3022**, Appendix D, "Documentation of VV&A for Models and Simulations," January 28, 2008 [2]
- (U) **NGIC 4.0 Documentation Templates**, January 30, 2026

UNCLASSIFIED		
Version	Date	Changes
1.0	2026-03-05	Initial release. Accreditation assessment for T-150 UAS AFSIM Plugin Suite v2.0 covering all six development phases.
		UNCLASSIFIED

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1 (U) Executive Summary

(U) This Accreditation Report documents the accreditation assessment and decision for the T-150/TRV-150 Heavy-Lift UAS AFSIM Simulation Model v2.0 (hereafter "the Model"). The Model comprises five custom C++ plugins totaling approximately 7,736 lines of code across 26 source files, developed for AFSIM version 2.2505. The plugins collectively model the flight dynamics, battery energy system, propulsion, Vortex Ring State (VRS), and slung load dynamics of the T-150 coaxial octocopter UAS.

(U) The accreditation assessment evaluated the Model against 13 requirements derived from the intended use of intelligence analysis of T-150 UAS operational capabilities. Verification and validation testing demonstrated that 12 of 13 requirements are fully met, with one requirement (R-009, Flight Endurance) partially met due to a calibration gap between simulated endurance (14.3 minutes) and manufacturer-specified endurance (36 minutes).

(U) **Overall Recommendation: CONDITIONAL ACCREDITATION** for intelligence analysis use, subject to the following key conditions:

- (U) Analysts must apply an endurance scaling factor when interpreting time-based results
- (U) Results should be used for comparative analysis (relative differences) rather than absolute performance predictions
- (U) VRS and slung load results should be treated as indicative of behavioral trends, not as predictive outputs
- (U) The Model should be recalibrated if T-150 flight test data becomes available

(U) The Model demonstrates physics-based fidelity sufficient for intelligence-grade analysis of T-150 operational capabilities, vulnerability assessment, and counter-UAS scenario development. Energy conservation has been validated to less than 0.01% error, subsystem interactions are correctly modeled, and behavioral trends align with established aerodynamic theory. The primary limitation is the absence of flight test data for direct validation, which is typical for threat system modeling based on open-source intelligence.

2 (U) Problem Statement

(U) The intelligence community requires a physics-based simulation of the T-150/TRV-150 Heavy-Lift UAS to answer analytical questions regarding the operational capabilities, performance envelope, and limitations of this heavy-lift multirotor platform. These questions support counter-UAS assessment, vulnerability analysis, and intelligence production.

(U) Native AFSIM kinematic movers lack the fidelity required to model rotor aerodynamics, battery depletion, motor thermal dynamics, Vortex Ring State, ground effect, and slung load interactions. A custom plugin suite was developed to provide parametric-level fidelity for these physics domains.

(U) The intended use of the simulation is intelligence analysis, specifically:

- (U) Assessing T-150 operational capabilities and performance boundaries
- (U) Comparative flight pattern efficiency analysis
- (U) Vulnerability assessment (VRS entry conditions, battery depletion profiles, thermal limits)
- (U) Payload impact analysis on flight performance
- (U) Counter-UAS (C-UAS) scenario development and evaluation

3 (U) M&S Requirements and Acceptability Criteria

3.1 (U) Requirements Traceability with Final Assessment

Table 2: (U) Requirements Final Assessment

UNCLASSIFIED				
Req ID	Description	Acceptability Criteria	Met?	Assessment Summary
R-001	Battery SOC accuracy	SOC tracking error $< 1\%$	**YES**	Energy conservation $< 0.01\%$ error in Test 09
R-002	Voltage sag model	Terminal voltage follows Li-Ion curve	**YES**	SOC-dependent resistance produces correct sag (Test 06)
R-003	Motor thermal dynamics	Stable equilibrium; no runaway	**YES**	Stable equilibrium under sustained load (Test 05)
R-004	Position hold stability	Drift < 0.5 m/s in hover	**YES**	PID controller maintains hover ≈ 0 m/s drift (Test 01)
R-005	Velocity tracking	Error < 1 m/s during cruise	**YES**	Achieves commanded velocities within spec (Test 02)
R-006	Numerical stability	No NaN, Inf, or divergence	**YES**	No instabilities under extreme inputs (Test 08)
R-007	VRS detection	Correct state transitions	**YES**	Correct transitions based on descent rate and airspeed (Test 03)
R-008	Ground effect	Thrust augmentation near ground	**YES**	Cheeseman-Bennett model produces correct increase (Test 07)
R-009	Flight endurance	Within 20% of manufacturer spec	**PARTIAL**	14.3 min vs. 36 min; behavioral trends correct; scaling required
R-010	Slung load dynamics	Correct pendulum frequency/damping	**YES**	Frequency consistent with cable length parameters (Test 10)
R-011	Attitude recovery	Recovery within 5 s from disturbance	**YES**	PID recovers from disturbances within specification (Test 04)
R-012	Cruise speed	Achieves 30 m/s	**YES**	30 m/s achieved and sustained (Test 02)
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Table 2: (U) Requirements Final Assessment (Continued)

UNCLASSIFIED				
Req ID	Description	Acceptability Criteria	Met?	Assessment Summary
R-013	Max payload capacity	Correctly limits to 68 kg	**YES**	68 kg modeled; performance degradation consistent (Test 11)
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3.2 (U) Requirements Assessment Summary

- (U) **Requirements Fully Met**: 12 of 13 (92.3%)
- (U) **Requirements Partially Met**: 1 of 13 (7.7%) – R-009 (Flight Endurance)
- (U) **Requirements Not Met**: 0 of 13 (0.0%)

(U) The single partially-met requirement (R-009) is attributable to a calibration gap in the power consumption model rather than a fundamental modeling error. The behavioral trend (monotonic SOC decline with correct voltage sag) is correct; the absolute rate of energy consumption is higher than the real system, likely due to conservative efficiency assumptions. This gap is well-documented and can be addressed through analyst-applied scaling factors.

4 (U) M&S Assumptions, Capabilities, Limitations, and Risks/Impacts

4.1 (U) Key Assumptions

1. (U) **Rigid body assumption**: The airframe is modeled as a rigid body. Structural flex, vibration modes, and aeroelastic effects are not represented.
2. (U) **Quasi-steady aerodynamics**: Rotor aerodynamics use momentum theory with quasi-steady assumptions.
3. (U) **Uniform inflow**: Each rotor is modeled with uniform inflow. BEMT radial variation is not implemented.
4. (U) **Estimated inertia tensor**: Moments of inertia are estimated from geometric assumptions and mass distribution.
5. (U) **Standard atmosphere**: Air density varies with altitude per ISA model.
6. (U) **Idealized battery cells**: All battery cells are assumed identical.
7. (U) **No wind turbulence**: The current model does not include atmospheric turbulence or gust modeling.
8. (U) **Open-source parameters**: All model parameters are derived from OSINT.

4.2 (U) Model Capabilities

- (U) **6DOF flight dynamics**: Full six-degree-of-freedom rigid body physics with Newton-Euler integration, PID attitude and velocity control, and waypoint navigation
- (U) **Battery energy modeling**: Coulomb counting SOC tracking, SOC-dependent OCV, load-dependent internal resistance and voltage sag, low-battery warnings
- (U) **Propulsion modeling**: Per-motor RPM and thrust calculation, motor thermal dynamics with convective cooling, dynamic power from flight state
- (U) **VRS detection**: State machine detecting VRS entry conditions with configurable thresholds
- (U) **Ground effect**: Cheeseman-Bennett thrust augmentation model
- (U) **Slung load dynamics**: Spherical pendulum model with force feedback through AuxData
- (U) **Inter-plugin communication**: Loosely-coupled blackboard architecture via AuxData

4.3 (U) Model Limitations

1. (U) **Endurance calibration gap**: Simulated endurance (14.3 min) is ≈40% of manufacturer specification (36 min). All time-based results require scaling.
2. (U) **No flight test validation**: No T-150 flight test data available for direct comparison.
3. (U) **Developer self-testing**: V&V performed by development team; no independent V&V agent.

4. (U) **Estimated parameters**: Several parameters estimated from class-level data.
5. (U) **Informational subsystems**: VRS and slung load are informational – they report state but do not dynamically alter flight trajectory.
6. (U) **No degraded-mode modeling**: Motor failure, partial power loss, GPS denial not modeled.
7. (U) **Single environmental condition**: ISA standard conditions only.

4.4 (U) Accreditation-Specific Risk Assessment

Table 3: (U) Accreditation Risk Assessment

UNCLASSIFIED		
Risk Category	Risk Level	Justification
Relative performance comparisons	**LOW**	Physics-based trends correct; comparative analysis well-supported
Absolute performance predictions	**MEDIUM**	Endurance gap introduces quantitative uncertainty
Time-critical mission analysis	**HIGH**	Without scaling, time-based predictions underestimate mission duration
Behavioral trend analysis	**LOW**	Physics-based models produce correct qualitative behavior
C-UAS engagement analysis	**LOW-MED**	Flight dynamics sufficient for geometry analysis; signatures not modeled
		UNCLASSIFIED

5 (U) Accreditation Assessment

5.1 (U) Assessment Summary

(U) The accreditation assessment evaluated the T-150 UAS AFSIM Plugin Suite v2.0 against 13 model requirements derived from the intended use of intelligence analysis. The assessment considered V&V test results, documentation completeness, configuration management practices, and overall suitability.

(U) **Result**: 12 of 13 requirements fully met. One requirement (R-009) partially met with documented calibration gap. The Model demonstrates sufficient fidelity for intelligence analysis when used with appropriate analyst awareness of its limitations.

5.2 (U) Strengths

1. (U) **Physics-based modeling provides correct behavioral trends**: Momentum theory, Newton-Euler dynamics, Coulomb counting, and first-principles thermal modeling ensure physically consistent results. Energy conservation validated to $<0.01\%$ error.
2. (U) **Energy conservation validated to high precision**: Test 09 demonstrated $<0.01\%$ energy balance error over full discharge cycles.
3. (U) **Multiple subsystem interactions correctly modeled**: The loosely-coupled plugin architecture successfully models interactions between all five subsystems.
4. (U) **Configuration management via git provides traceability**: All source code, scenarios, and documentation maintained with full commit history.
5. (U) **Comprehensive documentation suite**: ADD v3.0, UAM v3.0, ICD v1.0, and TDD provide full traceability.
6. (U) **Modular architecture enables incremental confidence**: Each plugin independently verifiable. Six-phase development allowed progressive confidence building.
7. (U) **Boundary condition testing demonstrates robustness**: No numerical instabilities under extreme conditions.

5.3 (U) Weaknesses

1. (U) **No flight test data for direct validation**: Validation relies on specifications, literature, and physics-based consistency. Common for threat system modeling.
2. (U) **Endurance gap requires analyst awareness**: 2.5x discrepancy means time-based outputs cannot be used at face value.
3. (U) **Developer self-testing**: Risk of confirmation bias in test design and interpretation.
4. (U) **Some estimated parameters**: Inertia, thermal mass, and resistance coefficients estimated from class-level data.
5. (U) **VRS and slung load are informational only**: Report state but do not impose full aerodynamic coupling.

5.4 (U) Risk Assessment

(U) Overall risk for intended use of intelligence analysis: **LOW to MEDIUM**.

- (U) **LOW risk** for comparative analysis and behavioral trend identification (primary intended use)
- (U) **MEDIUM risk** for quantitative performance predictions (require scaling factors)
- (U) **HIGH risk** for time-critical predictions without endurance scaling (mitigated by documented conditions)

6 (U) Accreditation Recommendations

6.1 (U) Recommendation

(U) **CONDITIONAL ACCREDITATION**

(U) The T-150/TRV-150 Heavy-Lift UAS AFSIM Simulation Model v2.0 is recommended for conditional accreditation for use in intelligence analysis of T-150 UAS operational capabilities. The conditions in Section 5.2 must be satisfied for all analytical uses.

6.2 (U) Conditions of Accreditation

(U) **(a) Endurance Scaling Requirement**: Analysts must apply an endurance scaling factor of approximately 2.5x when interpreting time-based simulation results. Simulated flight durations should be multiplied by 2.5 to estimate real-world endurance. This factor must be documented in all analytical products referencing simulation-derived time or range values.

(U) **(b) Comparative Analysis Preference**: Results should be used for comparative analysis (relative differences between configurations) rather than absolute performance predictions. The Model's strength lies in correctly capturing the direction and relative magnitude of performance changes.

(U) **(c) VRS and Slung Load Interpretation**: VRS detection and slung load dynamics results should be treated as indicative of behavioral trends, not as predictive of specific outcomes. The VRS processor identifies conditions consistent with VRS entry but does not impose full aerodynamic coupling.

(U) **(d) Recalibration on Data Availability**: The Model should be recalibrated if T-150 flight test data, detailed motor efficiency curves, or measured battery discharge profiles become available. Priority calibration targets: motor efficiency vs. RPM curves, battery internal resistance vs. SOC and temperature, and propeller thrust coefficient vs. advance ratio.

6.3 (U) Approved Uses

(U) The Model is approved for the following analytical applications, subject to the conditions in Section 5.2:

- (U) **Intelligence analysis** of T-150 UAS operational capabilities, performance envelope, and system limitations
- (U) **Comparative flight pattern efficiency analysis**, including mission profiles, waypoint optimization, and altitude/speed trade-offs
- (U) **Vulnerability assessment**, including VRS susceptibility, battery depletion profiles, motor thermal limits, and operational boundary identification
- (U) **Payload impact analysis**, evaluating payload mass effects on performance, endurance, and maneuverability
- (U) **Counter-UAS (C-UAS) scenario development**, providing representative UAS flight dynamics for detection, tracking, and engagement analysis

- (U) **Concept of Operations (CONOPS) evaluation**, assessing tactical employment concepts for heavy-lift UAS operations
- (U) **Training and familiarization**, demonstrating heavy-lift UAS flight characteristics

6.4 (U) Prohibited Uses

(U) The Model is **not approved** for:

- (U) **Real-time flight control** or any application involving actual aircraft operations
- (U) **Safety-critical decision making** where model inaccuracy could result in loss of life or equipment
- (U) **Acquisition decisions** based solely on model output without independent data sources
- (U) **Absolute performance guarantees** without application of the endurance scaling factor and documented uncertainty bounds
- (U) **Prediction of specific failure modes** beyond the behavioral indicators modeled
- (U) **Certification or airworthiness assessment** of any UAS system

7 (U) Key Participants

Table 4: (U) Key Participants

UNCLASSIFIED	
Role	Name/Organization
Accreditation Authority	[PLACEHOLDER - Name, Title, Organization]
M&S Proponent	[PLACEHOLDER - Name, Title, Organization]
V&V Agent	[PLACEHOLDER - Name, Title, Organization]
SME (UAS)	[PLACEHOLDER - Name, Title, Organization]
SME (Aerodynamics)	[PLACEHOLDER - Name, Title, Organization]
Model Developer	[PLACEHOLDER - Name, Title, Organization]
Configuration Manager	[PLACEHOLDER - Name, Title, Organization]
Analyst Representative	[PLACEHOLDER - Name, Title, Organization]
	UNCLASSIFIED

8 (U) Actual Accreditation Resources Expended

Table 5: (U) Resources Expended

UNCLASSIFIED			
Resource Category	Planned	Actual	Notes
Calendar Duration	[PLACEHOLDER]	∼57 days	Development start to accreditation package
Active Development Hours	[PLACEHOLDER]	∼20 hours	AI-assisted development reduced effort
V&V Testing Hours	[PLACEHOLDER]	[PLACEHOLDER]	Includes scenario development, execution, analysis
Documentation Hours	[PLACEHOLDER]	[PLACEHOLDER]	AI-assisted documentation reduced effort
Review and Assessment Hours	[PLACEHOLDER]	[PLACEHOLDER]	SME review, accreditation meetings
Total Labor (person-hours)	[PLACEHOLDER]	[PLACEHOLDER]	Sum of all categories
			UNCLASSIFIED

(U) **Note**: AI-assisted development using Claude Code CLI substantially reduced both code development and documentation timelines. Refer to the Impact Statement [12] for detailed discussion.

9 (U) Accreditation Lessons Learned

9.1 (U) AI-Assisted Development

(U) AI-assisted development significantly accelerated model creation, producing over 7,700 lines of C++ code and comprehensive VV&A documentation within approximately 20 hours. However, this approach required human calibration oversight – the AI-generated code was architecturally sound but required manual intervention for motor efficiency calibration.

(U) **Recommendation**: AI-assisted M&S development should include explicit calibration review gates where SMEs validate key performance parameters before integration testing.

9.2 (U) AFSIM Plugin Architecture

(U) The AFSIM plugin architecture enabled modular V&V. The AuxData blackboard communication pattern simplified interface testing. This modularity directly supported the six-phase iterative development approach.

(U) **Recommendation**: Future AFSIM-based M&S efforts should adopt modular plugin architecture with AuxData communication to facilitate incremental V&V.

9.3 (U) Open-Source Intelligence as Basis for Threat Modeling

(U) OSINT provided sufficient data for parametric-level fidelity. The AAIB report, patent filing, motor specs, and product data collectively provided approximately 65 system parameters.

(U) **Recommendation**: OSINT-based threat models should document source and confidence level for every parameter.

9.4 (U) Physics-Based Validation

(U) Physics-based models provide inherent validation through energy conservation and dimensional analysis. The $<0.01\%$ energy accounting accuracy provides confidence in the underlying physics engine even without flight test data.

(U) **Recommendation**: For threat system models where flight test data is unavailable, physics-based consistency checks should be a primary validation technique.

9.5 (U) Iterative Development and Progressive Confidence

(U) The six-phase approach allowed progressive confidence building. Each phase added capability while regression testing preserved previously validated behavior.

(U) **Recommendation**: Complex M&S efforts should adopt iterative development with defined phase gates.

9.6 (U) Configuration Management

(U) Git and GitHub provided excellent traceability. Every code change is linked to a specific commit, branch, and pull request. The development history is fully auditable.

(U) **Recommendation**: All M&S efforts subject to VV&A should use version control with branch-based workflows.

A (U) M&S Description

Table 6: (U) M&S Components

UNCLASSIFIED				
Plugin	Library	Registered Type	LOC	Purpose
Rotorcraft 6DOF Mover	rotorcraft_6dof	WSF_ ROTORCRAFT_ 6DOF_MOVER	2,939	Physics-based 6DOF flight dynamics with PID control
Battery Processor	t150_battery	T150_BATTERY_ PROCESSOR	1,139	Li-Ion battery SOC tracking and voltage sag
Propulsion Processor	t150_propulsion	T150_PROPULSION_ PROCESSOR	1,281	Motor power, thermal dynamics, RPM, ground effect
VRS Processor	t150_vrs	T150_VRS_ PROCESSOR	697	VRS detection and state machine
Slung Load Processor	t150_slung_load	T150_SLUNG_ LOAD_PROCESSOR	825	Cable-suspended payload pendulum dynamics
Total			**7,736**	
				UNCLASSIFIED

(U) The plugins communicate through the AFSIM AuxData key-value store. For full architecture description, refer to the ADD v3.0 [7].

B (U) Requirements Traceability Matrix (Final)

Table 7: (U) Final Requirements Traceability

UNCLASSIFIED				
Req ID	V&V Plan Ref	Test ID	Result	Status
R-001	Sec 3.1	Test 09	**PASS** ($<$ 0.01% error)	ACCEPTED
R-002	Sec 3.2	Test 06	**PASS**	ACCEPTED
R-003	Sec 3.3	Test 05	**PASS**	ACCEPTED
R-004	Sec 3.4	Test 01	**PASS** (drift $\sim$ 0 m/s)	ACCEPTED
R-005	Sec 3.5	Test 02	**PASS** (error $<$ 1 m/s)	ACCEPTED
R-006	Sec 3.6	Test 08	**PASS** (no NaN)	ACCEPTED
R-007	Sec 3.7	Test 03	**PASS**	ACCEPTED
R-008	Sec 3.8	Test 07	**PASS**	ACCEPTED
R-009	Sec 3.9	Test 12	PARTIAL (14.3 vs 36 min)	ACCEPTED WITH CONDITIONS
R-010	Sec 3.10	Test 10	**PASS**	ACCEPTED
R-011	Sec 3.11	Test 04	**PASS**	ACCEPTED
R-012	Sec 3.12	Test 02	**PASS** (30 m/s achieved)	ACCEPTED
R-013	Sec 3.13	Test 11	**PASS** (68 kg modeled)	ACCEPTED
				UNCLASSIFIED

C (U) Basis of Comparison

Table 8: (U) Basis of Comparison

UNCLASSIFIED		
Source	Type	Parameters Validated
AAIB Investigation Report [4]	Accident investigation	MTOW, thrust, configuration
US Patent US20240262499A1 [5]	Patent filing	Battery configuration, voltage, capacity
T-Motor U15II KV100 Specs [6]	Manufacturer data	Motor KV, max power, dimensions
Cheeseman & Bennett (1955) [8]	Published research	Ground effect thrust augmentation
Johnson (1980) [9]	Textbook	Helicopter rotor aerodynamics, VRS
Leishman (2006) [10]	Textbook	Rotorcraft aerodynamics, momentum theory
TRV-150 Product Data [11]	Manufacturer marketing	Cruise speed, max payload, endurance
		UNCLASSIFIED

D (U) References

(U) [1] DoD Instruction 5000.61, "DoD M&S VV&A," September 17, 2024.

(U) [2] MIL-STD-3022, "Documentation of VV&A for Models and Simulations," January 28, 2008.

(U) [3] Army Directive 2024-02: Software Development, 2024.

(U) [4] AAIB, "Investigation to Malloy Aeronautics T-150, South Scarle, Nottinghamshire," 2023.

(U) [5] Malloy Aeronautics, "Battery Pack Topology for UAV," US Patent US20240262499A1, 2024.

(U) [6] T-Motor, "U15II KV100 Motor Specifications," 2024.

(U) [7] T-150 UAS AFSIM Plugin Suite, "ADD v3.0," March 5, 2026.

(U) [8] I.C. Cheeseman and W.E. Bennett, "The Effect of Ground on a Helicopter Rotor," ARC R&M 3021, 1955.

(U) [9] W. Johnson, "Helicopter Theory," Princeton University Press, 1980.

(U) [10] J.G. Leishman, "Principles of Helicopter Aerodynamics," 2nd ed., Cambridge, 2006.

(U) [11] SURVICE / Malloy Aeronautics, "TRV-150 Heavy-Lift UAS Product Data Sheet," 2024.

(U) [12] NGIC, "Impact of AI-Assisted Development on M&S: T-150 UAS Case Study," March 4, 2026.

E (U) Acronyms

Table 9: (U) Acronyms

UNCLASSIFIED	
Acronym	Definition
ADD	Architecture Description Document
AFSIM	Advanced Framework for Simulation, Integration, and Modeling
AGL	Above Ground Level
BEMT	Blade Element Momentum Theory
CG	Center of Gravity
C-UAS	Counter-Unmanned Aerial System
CONOPS	Concept of Operations
DLL	Dynamic Link Library
DoD	Department of Defense
DoDI	DoD Instruction
6DOF	Six Degrees of Freedom
ESC	Electronic Speed Controller
GE	Ground Effect
ICD	Interface Control Document
IGE	In Ground Effect
ISA	International Standard Atmosphere
KV	Motor velocity constant (RPM per Volt)
M&S	Modeling and Simulation
MTOW	Maximum Takeoff Weight
NaN	Not a Number
NGIC	National Ground Intelligence Center
OCV	Open-Circuit Voltage
OGE	Out of Ground Effect
OSINT	Open-Source Intelligence
PID	Proportional-Integral-Derivative
RPM	Revolutions Per Minute
SME	Subject Matter Expert
SOC	State of Charge
TDD	Test Design Document
	UNCLASSIFIED

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Table 9: (U) Acronyms (Continued)

UNCLASSIFIED	
Acronym	Definition
UAM	User and Analyst Manual
UAS	Unmanned Aerial System
V&V	Verification and Validation
VRS	Vortex Ring State
VV&A	Verification, Validation, and Accreditation
	UNCLASSIFIED

F (U) Glossary

Table 10: (U) Glossary

UNCLASSIFIED	
Term	Definition
Accreditation	Official certification that a model or simulation is acceptable for use for a specific purpose. Per DoDI 5000.61.
AuxData	AFSIM's platform-level key-value attribute store used for inter-processor communication.
Blackboard architecture	A software pattern where components communicate by reading and writing to a shared data store.
Calibration gap	A quantitative discrepancy between model output and reference data that is understood and documented.
Conditional accreditation	An accreditation decision approving use subject to specified conditions and limitations.
Coulomb counting	A battery SOC estimation method that integrates current over time.
Ground effect	The increase in rotor thrust efficiency when operating near the ground.
Momentum theory	An analytical framework for rotor performance based on conservation of momentum.
Parametric-level fidelity	A level where key system parameters are represented with values traceable to source data.
Vortex Ring State	A hazardous condition when a rotorcraft descends into its own downwash.
	UNCLASSIFIED

G (U) Accreditation Plan (Reference)

(U) The Accreditation Plan for the T-150 UAS AFSIM Simulation Model v2.0 is maintained as a separate document. Refer to the Accreditation Plan v1.0 for accreditation strategy, required evidence, assessment criteria, and decision framework.

H (U) V&V Report (Reference)

(U) The V&V Report for the T-150 UAS AFSIM Simulation Model v2.0 is maintained as a separate document. The report contains detailed test procedures, raw results, pass/fail determinations, and analysis supporting the requirements assessment in Section 2.

(U) Signature Block

Table 11: (U) Accreditation Authority Signature

UNCLASSIFIED	
Field	Value
Accreditation Authority	[PLACEHOLDER - Name, Title, Organization]
Signature	[PLACEHOLDER]
Date	[PLACEHOLDER]
Accreditation Decision	**CONDITIONAL ACCREDITATION**
Effective Date	[PLACEHOLDER]
Expiration / Review Date	[PLACEHOLDER - Typically 2 years from effective date]
	UNCLASSIFIED

Table 12: (U) V&V Agent Signature

UNCLASSIFIED	
Field	Value
V&V Agent	[PLACEHOLDER - Name, Title, Organization]
Signature	[PLACEHOLDER]
Date	[PLACEHOLDER]
V&V Recommendation	Conditional Accreditation recommended based on V&V findings
	UNCLASSIFIED

Table 13: (U) M&S Proponent Signature

UNCLASSIFIED	
Field	Value
M&S Proponent	[PLACEHOLDER - Name, Title, Organization]
Signature	[PLACEHOLDER]
Date	[PLACEHOLDER]
Concurrence	[PLACEHOLDER]
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I (U) Acronyms

Citations